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palsr: Projected Actor Locations for spatial modeling of dyadic interactions between moving actors

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Software

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Summary

Many actors studied in the social sciences have no fixed geographic location. Armed groups roam across a theater of conflict, firms relocate, diplomats and international organizations shift their operational focus, and migrating populations move in response to events. For such *moving actors*, the location at which two of them interact is itself a meaningful, modelable outcome, but standard spatial-interaction tools assume actors sit at fixed points and so cannot be applied directly. The Projected Actor Location (PALS) method, introduced by Kim et al. (2023), fills this gap by estimating where a mobile actor effectively "is" at any point in time from the spatiotemporal history of its past interactions, and using these projected locations to model the distance between — and the probability of interaction among — pairs of actors.

`palsr` is an R package that implements the complete PALS workflow: constructing and validating dyadic event data, estimating the smoothing parameters by minimizing great-circle prediction error, projecting actor locations at arbitrary times, building the dyadic distance covariates used in downstream interaction models, and quantifying uncertainty through a nonparametric bootstrap with multiple-imputation (Rubin's Rules) pooling. Performance-critical kernels — the Haversine distance and the projection smoother — are implemented in C++ through `Rcpp` (Eddelbuettel 2013), so that parameter estimation, which repeatedly re-projects every actor across a grid of times, runs quickly even on data sets with thousands of events.

Statement of need

The empirical study of geopolitical interaction — alliances, conflict, cooperation, trade — is overwhelmingly *dyadic*: the unit of analysis is a pair of actors, and a central predictor of whether and how two actors interact is the distance between them (Gleditsch and Ward 2001; Ward et al. 2007). When actors are states with fixed capitals or centroids, this distance is trivial to compute. But a large and growing share of substantively important actors — rebel groups, militias, terrorist organizations, peacekeeping deployments, non-governmental organizations — are mobile, and treating them as fixed-location entities discards exactly the spatial dynamics that drive their interactions. Prior to PALS there was no general, estimable method for assigning time-varying locations to such actors in a way that is optimized for predicting their interactions.

PALS addresses this need, and the original article (Kim et al. 2023) demonstrates that PALS-based distances substantially improve the prediction of subnational conflict relative to naive location measures. However, that contribution was accompanied only by replication scripts specific to a single application. `palsr` turns the method into reusable, documented, and tested research software. It lets applied researchers in political science, conflict studies, economics, and geography apply PALS to their own dyadic event data without re-deriving the estimator, provides a principled bootstrap procedure for propagating estimation uncertainty into downstream models, and supplies a simulated example data set so that the full workflow can be learned and reproduced without access to restricted conflict data. To our knowledge it is the first general-purpose software implementation of projected-location modeling for moving actors.

Method

For a focal actor i at prediction time t, PALS forms a recency-weighted mean of the locations of i's own past events (the *focal* component) and a recency-weighted mean of the locations of events involving i's past interaction partners, or *alters* (the *alter* component). The projected location is a convex combination of the two,

$$g_i(t) = (1-\pi)\sum_e W_i(e)\, g(e) \;+\; \pi \sum_e W_k(e)\, g(e), \qquad \pi = \text{logistic}(\gamma + \eta\, v),$$

where event weights decay with age — governed by α for the focal actor and β for the alters — and the mixing weight π depends through γ and η on the focal actor's activity relative to its alters, summarized by the event-count statistic v. The four parameters are estimated by "marching forward" through time: every event is predicted using only events strictly preceding it, and parameters are chosen to minimize the mean Haversine distance between predicted and observed interaction locations. The package also exposes a parsimonious one-parameter variant that fixes $\pi = 0$ and estimates only α, which is fast and frequently competitive.

Key features

- **Validated data container** (`pal_events()`) for dyadic, time-stamped, georeferenced events.
- **Parameter estimation** (`estimate_pals()`) for both the full four-parameter and the reduced one-parameter models, via Haversine-error minimization with a compiled objective.
- **Projection** (`project_pal()`, `project_pals()`) of actor locations at any set of times.
- **Prediction and covariate construction** (`predict_event_locations()`, `pal_distance()`) for interaction locations and dyadic distances, including a log transform for conflict-style specifications.
- **Uncertainty quantification** (`bootstrap_pals()`, `pool_rubin()`) via a nonparametric event bootstrap and Rubin's Rules (Rubin 1987).
- **Reproducible example data** (`nigeria_sim`, `simulate_conflict_events()`), a vignette, and a full `testthat` suite that also checks the C++ kernels against pure-R reference implementations.

Example

```
library(palsr)
data(nigeria_sim)                          # 1,500 dyadic events, 25 mobile actors

fit <- estimate_pals(nigeria_sim, model = "one")    # estimate alpha
coef(fit)

# Project where each actor is on a given date.
project_pals(nigeria_sim, predict_time = as.Date("2015-01-01"), params = fit)
```

```r
# Build the dyadic distance covariate for an interaction model.
dyads <- data.frame(actor1 = "G01", actor2 = "G02", time = as.Date("2014-06-01"))
pal_distance(nigeria_sim, dyads, fit, transform = "log")

# Bootstrap uncertainty in the smoothing parameter.
summary(bootstrap_pals(nigeria_sim, R = 10, model = "one", seed = 1))
```


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